UNIVERSITY of HOUSTON | ECE

ECE 3337: Signals & Systems Analysis

Lecture 2: Basic Waveforms

Reference: Chapter 1
(Sections 1.3 – 1.4, Appendix B)

Please put away the gadgets & pay attention

Recap

Input Signal $\rightarrow$ **System** $\rightarrow$ Output Signal

Representing a signal is the first step to signal processing

Basic Signal Transformations:

$$\mathit{Shift}: y(t) = x(t - T)$$

$$\mathit{Scaling}: y(t) = x(at)$$

$$\text{Reversal}: y(t) = x(-t)$$

$$\text{Combined}: y(t) = x(at - b)$$

$$\boxed{\begin{array}{l} \text{If } |a| < 1, \text{ then } x(t) \text{ is expanded} \\ \text{If } |a| > 1, \text{ then } x(t) \text{ is compressed} \end{array}}$$

“PLAYBACK SPEED” ANALOGY

Recap: Combined Transformations

$$y(t) = x(at - b)$$

Think through your strategy before attempting problems!

$$x(t) \xrightarrow{\text{Time Scale}} x(at) \xrightarrow{\text{Shift}} x(a(t - \frac{b}{a})) \qquad \#1$$

$$x(t) \xrightarrow{\text{Shift}} x(t - b) \xrightarrow{\text{Time Scale}} x(at - b) \qquad \#2$$

ONLY THE ABOVE SEQUENCES OF TRANSFORMATIOS ARE VALID

HOW DO WE TELL THAT A STRATEGY IS VALID??

Today

We will explore basic waveform properties

- Even / Odd Symmetry
- Periodic and Non-periodic waveforms

We will introduce some common periodic and non-periodic waveforms

- Sinusoids
- Step function
- Ramp function

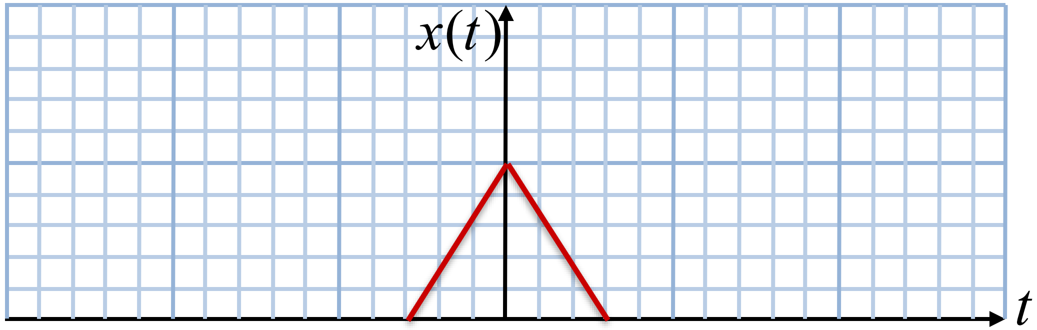
We will briefly review some math that we will need later

- Complex variables

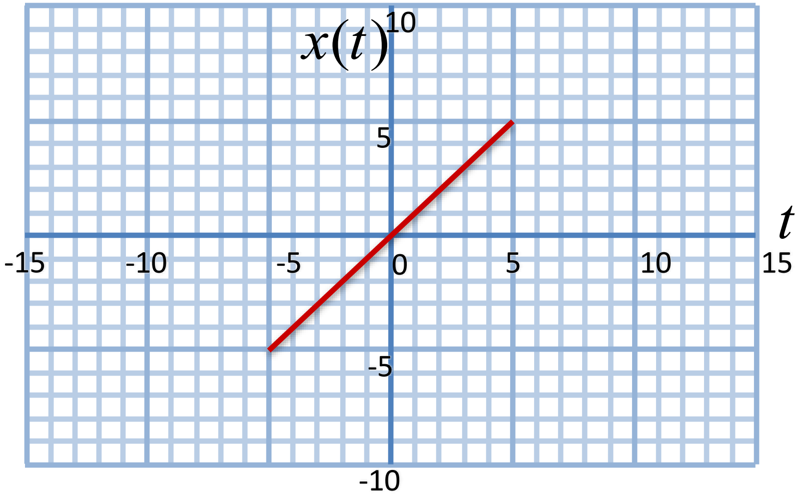
Symmetry

For some signals, $x(t)=x(-t)$

– This is a case of **even symmetry**



For some other signals, $x(t)=-x(-t)$



Odd symmetry

Question

What kinds of symmetry do the following signals exhibit?

$$x(t) = \sin(t)$$

$$x(t) = \cos(t)$$

$$x(t) = \tan(t)$$

$$x(t) = |t|$$

WHY DO WE CARE ABOUT SYMMETRY?

WHEN A SIGNAL HAS SYMMETRY, SOME CALCULATIONS CAN BECOME EASIER, AND WE CAN DEDUCE USEFUL PROPERTIES

Interesting Properties

For <u>any</u> signal, it is easy to verify that

$$x_e(t) = \frac{x(t) + x(-t)}{2}$$

is always even symmetric, because:

$$x_e(-t) = \frac{x(-t) + x(t)}{2} = x_e(t)$$

Now, verify that the following signal is always odd symmetric:

$$x_o(t) = \frac{x(t) - x(-t)}{2}$$

Cool Observation

$$x_e(t) = \frac{x(t) + x(-t)}{2} \qquad x_o(t) = \frac{x(t) - x(-t)}{2}$$

Easy to see that <u>any</u> signal can be represented as a sum of an even symmetric signal and an odd symmetric signal

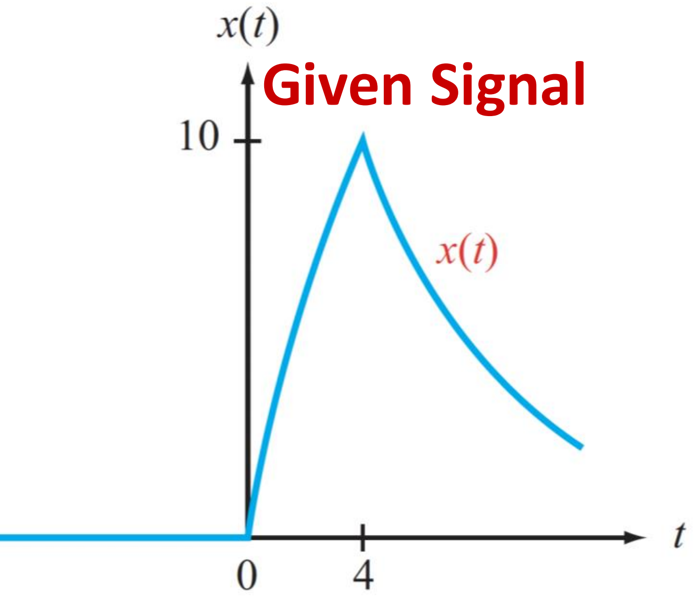
$$x(t) = x_e(t) + x_o(t)$$

This is a surprisingly big & powerful idea!

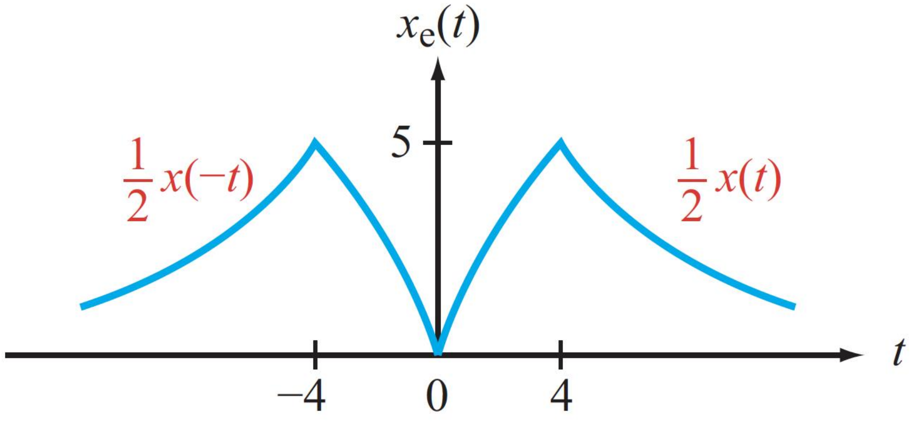
Representing complicated signals as a sum of simpler signals allows us to process them using a "divide and conquer" strategy

Example

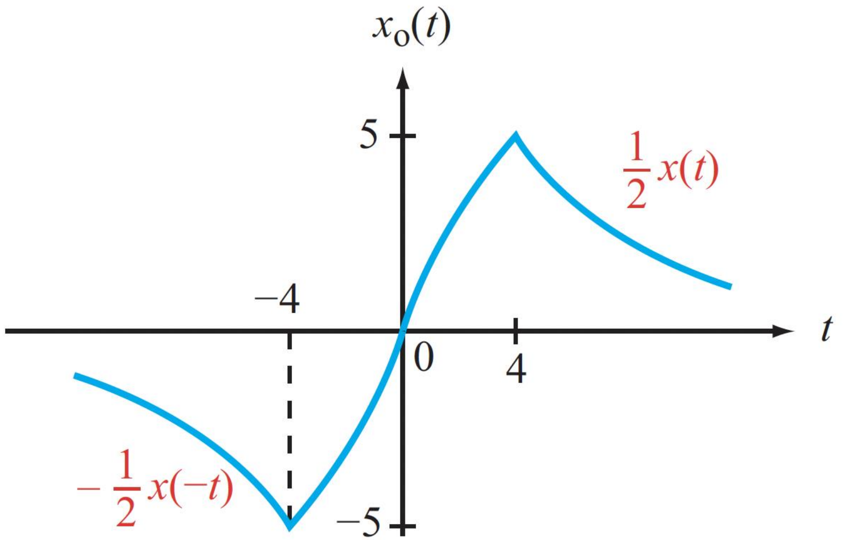
Given Signal



Even Component



Odd Component



$$x_e(t) = \frac{x(t) + x(-t)}{2}$$

$$x_o(t) = \frac{x(t) - x(-t)}{2}$$

Periodic Signals

Signals that keep on repeating, every T_0 seconds, forever

$$x(t) = x(t + nT_0)$$

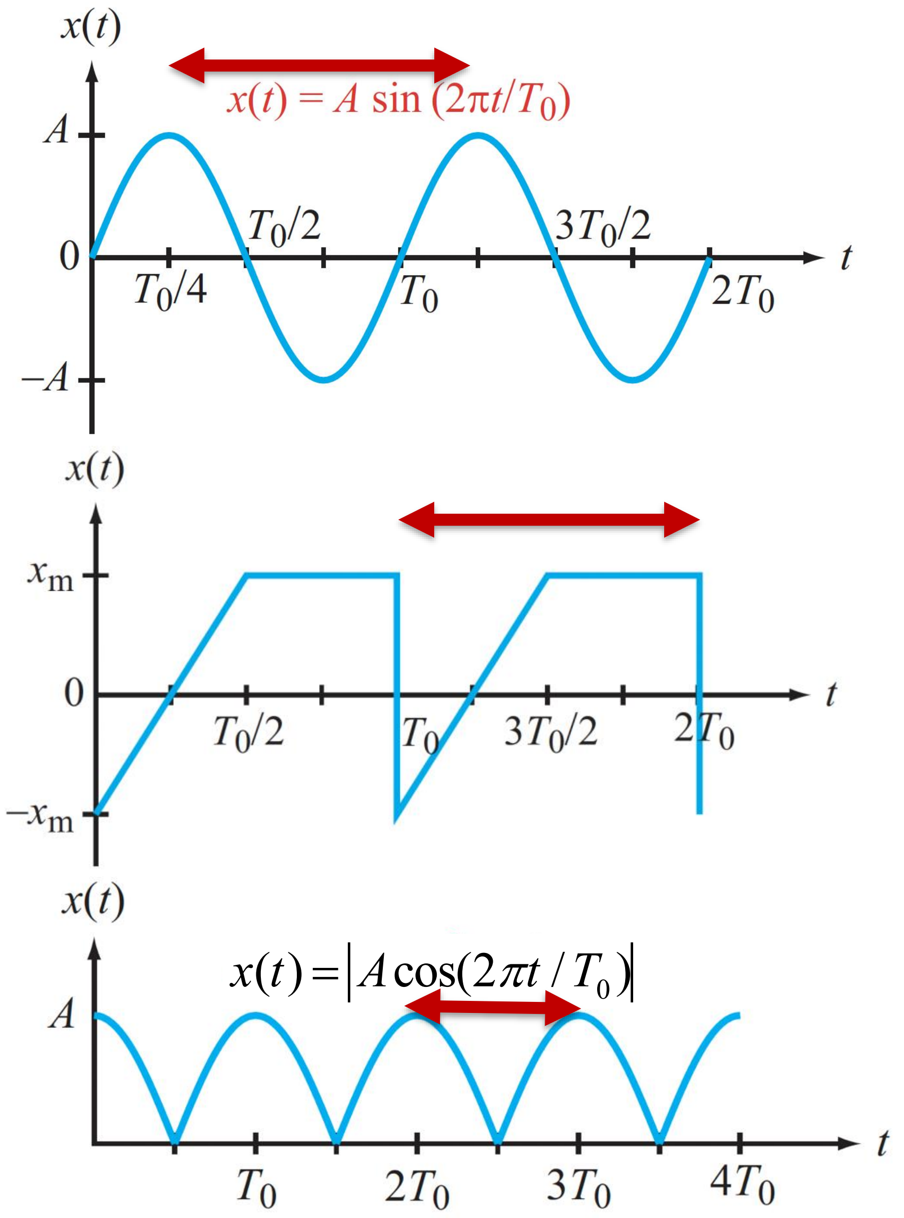
integer

Time Period
(seconds)

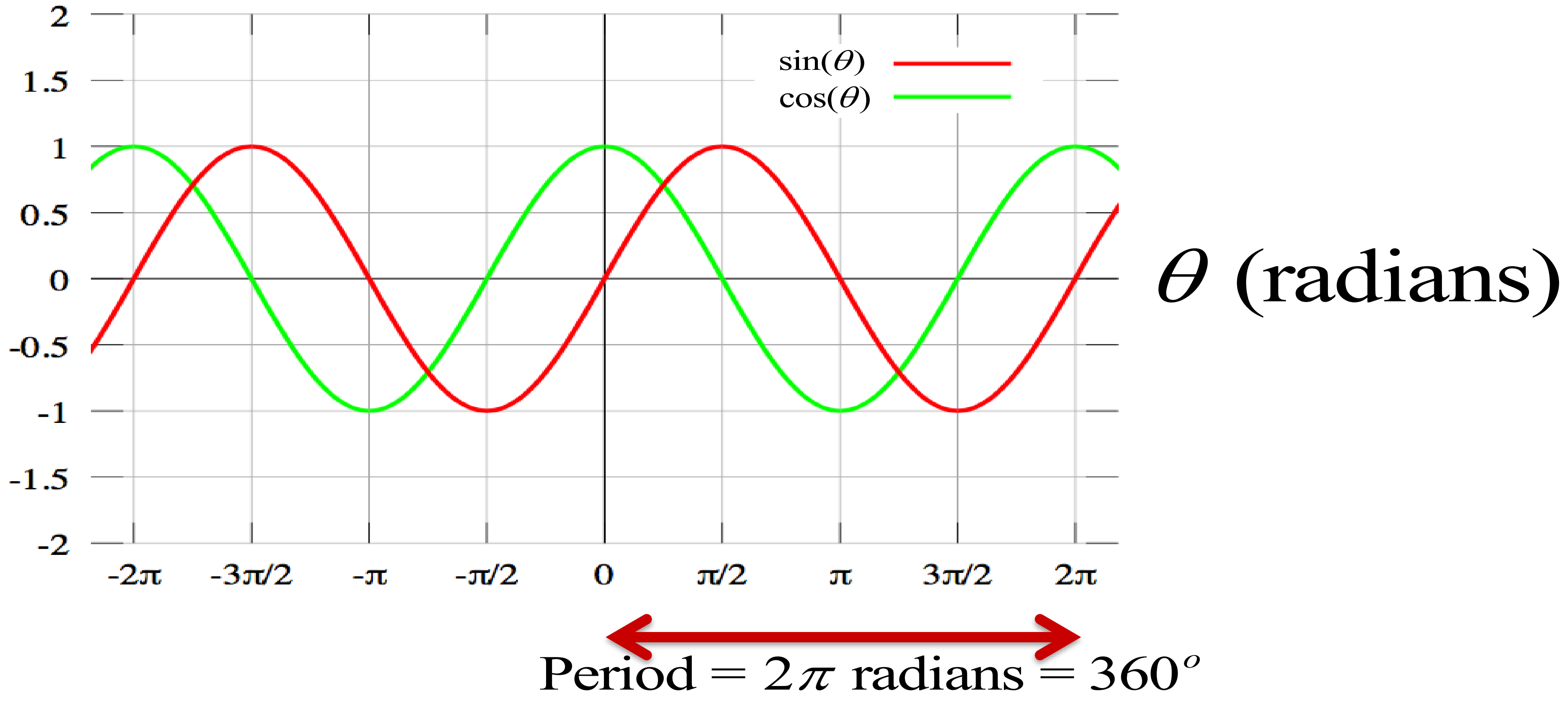
$$f_0 = \frac{1}{T_0}$$

Frequency
(repetitions per sec)

T_0 is the time between corresponding "landmark" points on successive periods



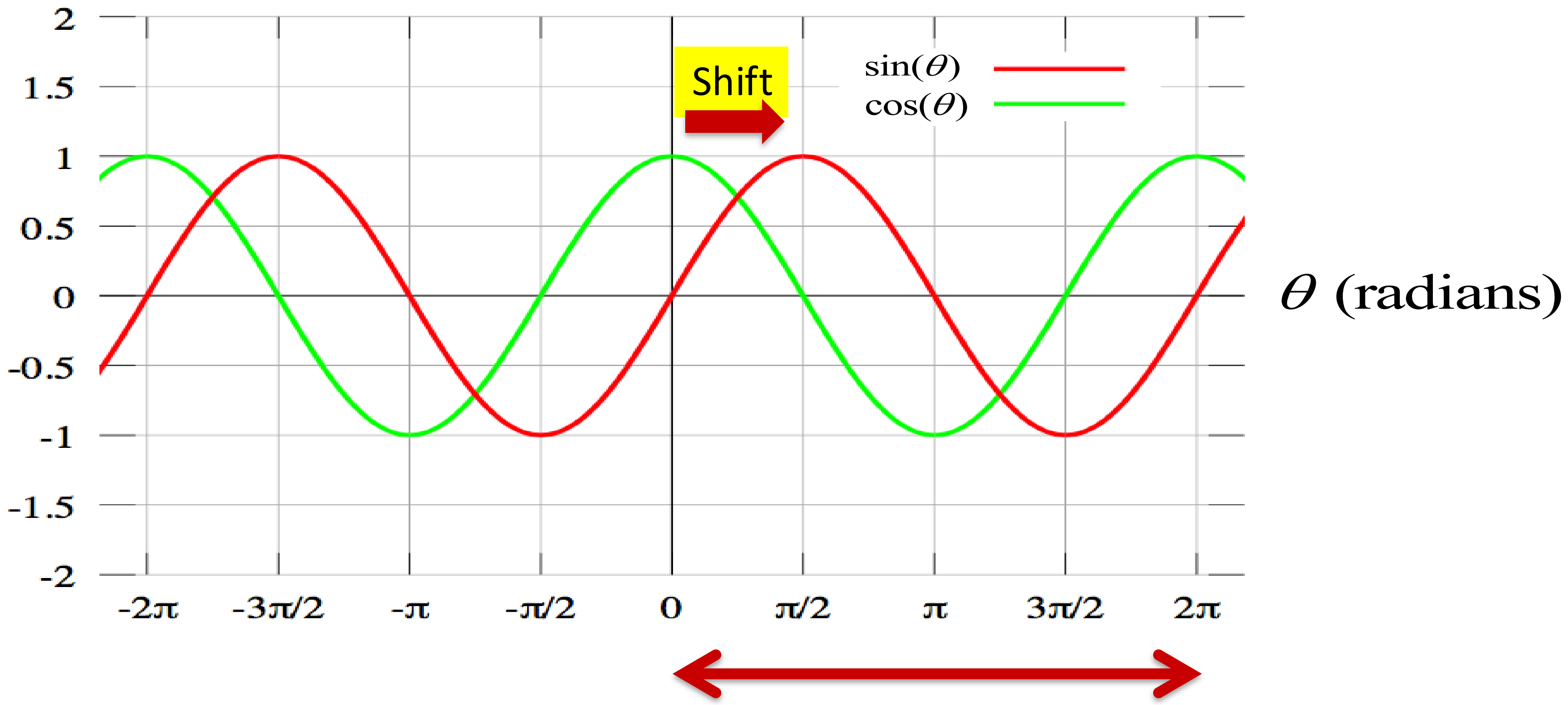
Sine and Cosine Functions



In Chapter 5, we will learn that any periodic waveform can be constructed as a weighted sum of sines and cosines

– **The big reason for studying them carefully now!**

Sine and Cosine



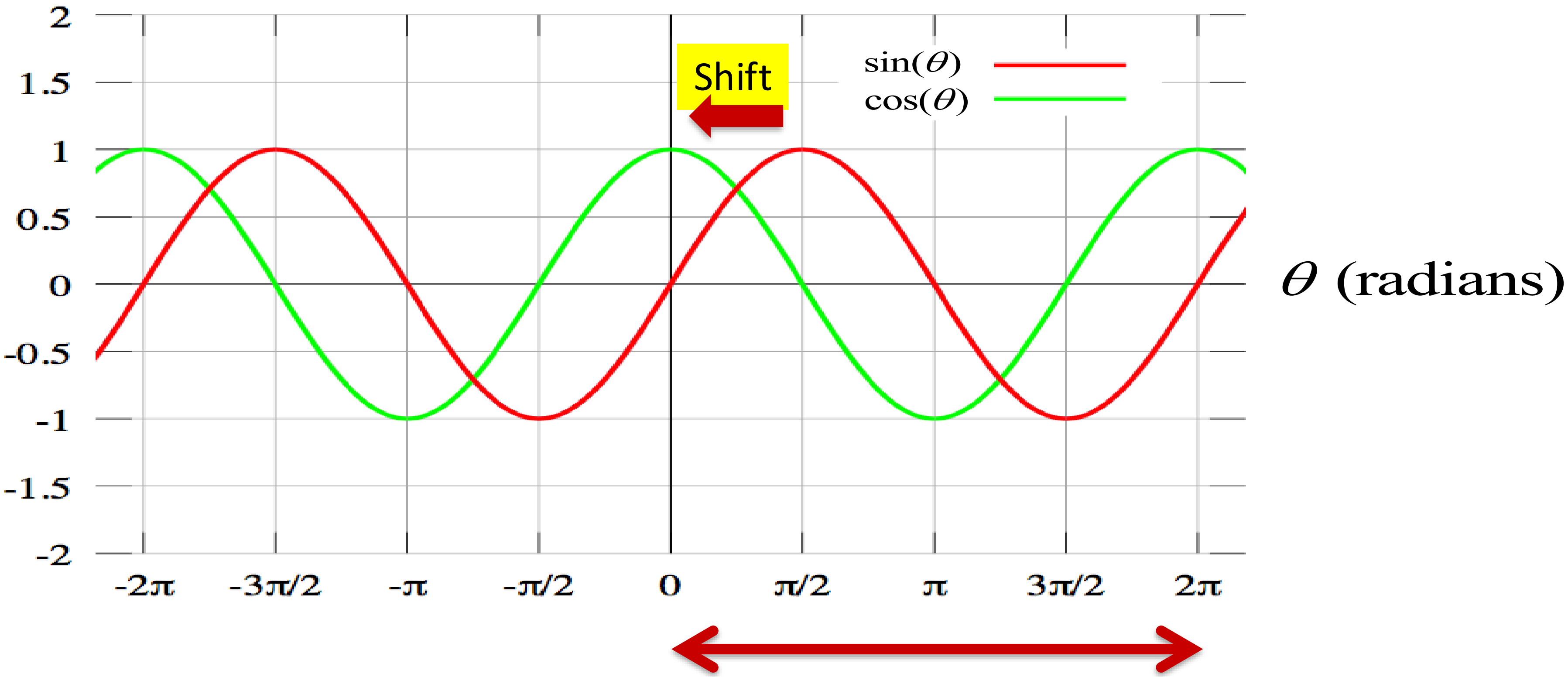
$$\text{Period} = 2\pi \text{ radians} = 360^o$$

$$\sin\theta = \cos(\theta - \frac{\pi}{2})$$

Remember:

$$y(t) = x(t - T)$$

Sine and Cosine



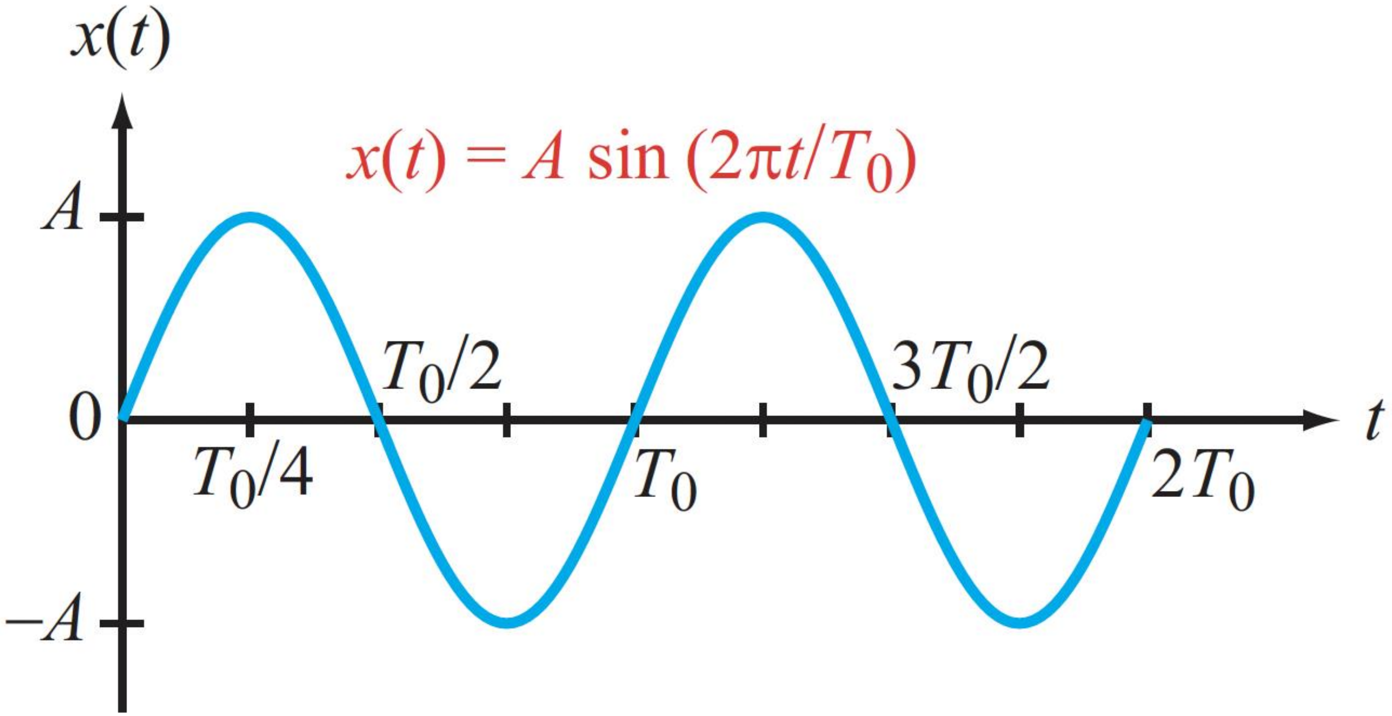
$$\text{Period} = 2\pi \text{ radians} = 360^o$$

$$\cos\theta = \sin(\theta + \frac{\pi}{2})$$

Remember:

$$y(t) = x(t - T)$$

Notation & Terms



$$x(t) = A\sin\left(\frac{2\pi t}{T_0} + \theta\right)$$

Phase Shift (radians)

Amplitude

Time Period

$$f_0 = \frac{1}{T_0}$$

Frequency
(per second)

$$\omega_0 = \frac{2\pi}{T_0}$$

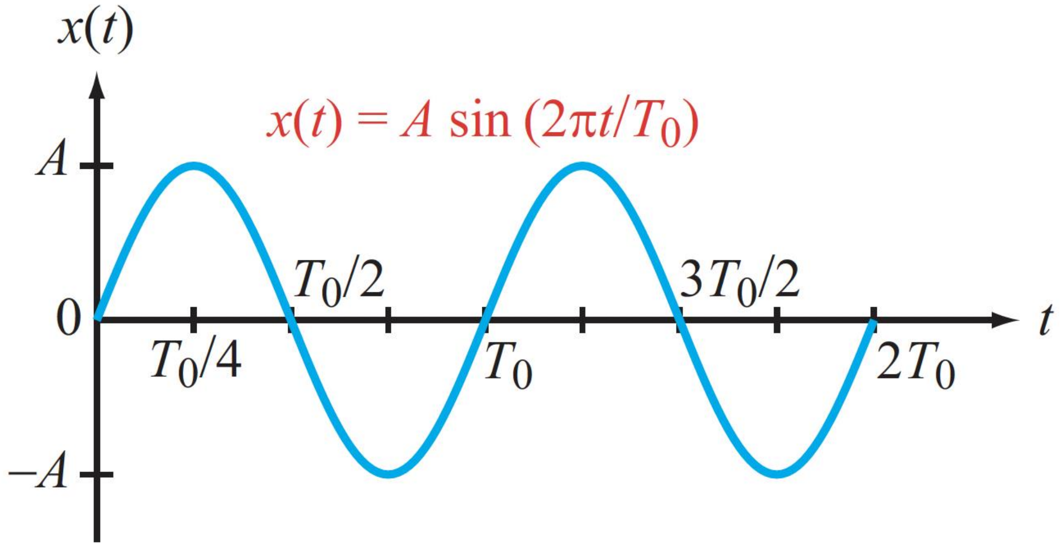
Angular frequency
(radians/second)

The Sinusoid

$$x(t) = A\sin\left(\frac{2\pi t}{T_0} + \theta\right)$$

$$= A\sin(\omega_0 t + \theta)$$

$$= A\sin(\omega_0 t + \omega_0 \tau)$$



$\tau =$ Time Delay equivalent to angle θ

$$\theta = \frac{2\pi\tau}{T_0}$$

When $\tau = T_0$, $\theta = 2\pi = 360^O$ ✓

When $\tau = \frac{T_0}{4}$, $\theta = \frac{\pi}{2} = 90^o$ ✓

Question: What does the case $\omega_0 = 0$ mean?

Euler's Equation

Euler's formula relates the sine and cosine functions with complex functions

$$e^{j\theta} = \cos(\theta) + j\sin(\theta)$$

$$e^{-j\theta} = \cos(\theta) - j\sin(\theta)$$

Euler's Number

Inverse Euler's Equation:

$$\cos(\theta) = \frac{1}{2}(e^{j\theta} + e^{-j\theta})$$

$$\sin(\theta) = \frac{1}{2j}(e^{j\theta} - e^{-j\theta})$$

THE MOST BEAUTIFUL EQUATION IN MATHEMATICS

Complex Variables

Complex variables are extremely useful for analyzing signals, circuits, and systems, even real-valued ones

- They are two-dimensional variables, so phase and magnitude information can be represented in a combined manner
- They make our formulas simpler and easier to understand
- They make our calculations shorter and more intuitive
- They provide a clean way to handle weird / tricky situations

It is important that you become familiar with this topic

- Importantly, learn to take advantage of complex representations, especially their two-dimensional nature
- Practice, and learn to avoid common mistakes

YouTube Video Lectures

- We will <u>briefly</u> go over some math background in class today
  - We will need to use complex variables throughout this course
- For your benefit, we have recorded a 3-part video lecture series on Complex Variables
  - Part 1: https://youtu.be/YxQ-yGu_QCo
  - Part 2: https://youtu.be/p0jqkdvHtt8
  - Part 3: https://youtu.be/yCSlUSc1aEg
  - The video lectures are just a review for ECE 3337
    - Not a substitute for a real math course!
- Several good textbooks available in PDF form online

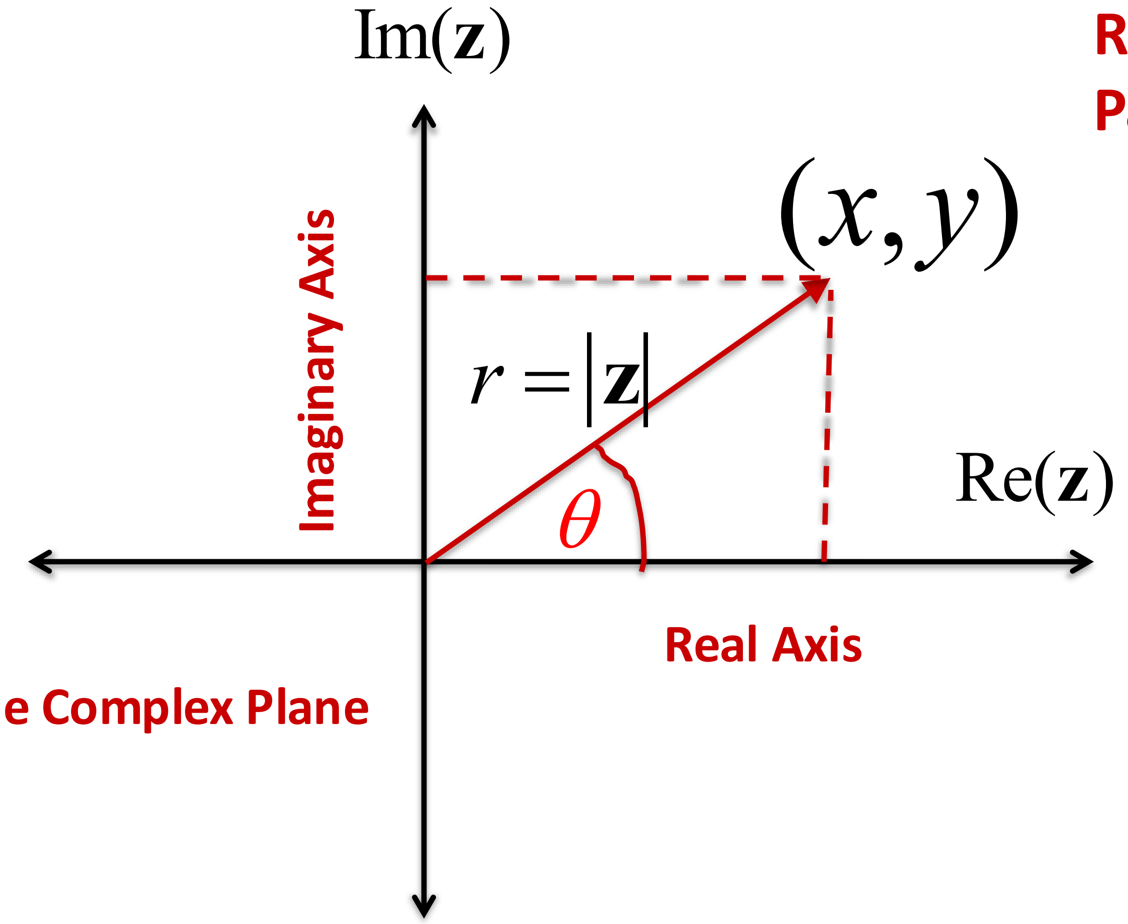
Complex Number Basics

Cartesian Form: $\mathbf{z} = x + jy, \quad \text{where } j = \sqrt{-1}$

Real Part

Imaginary Part

ECE folks like to use *i* for current



The Complex Plane

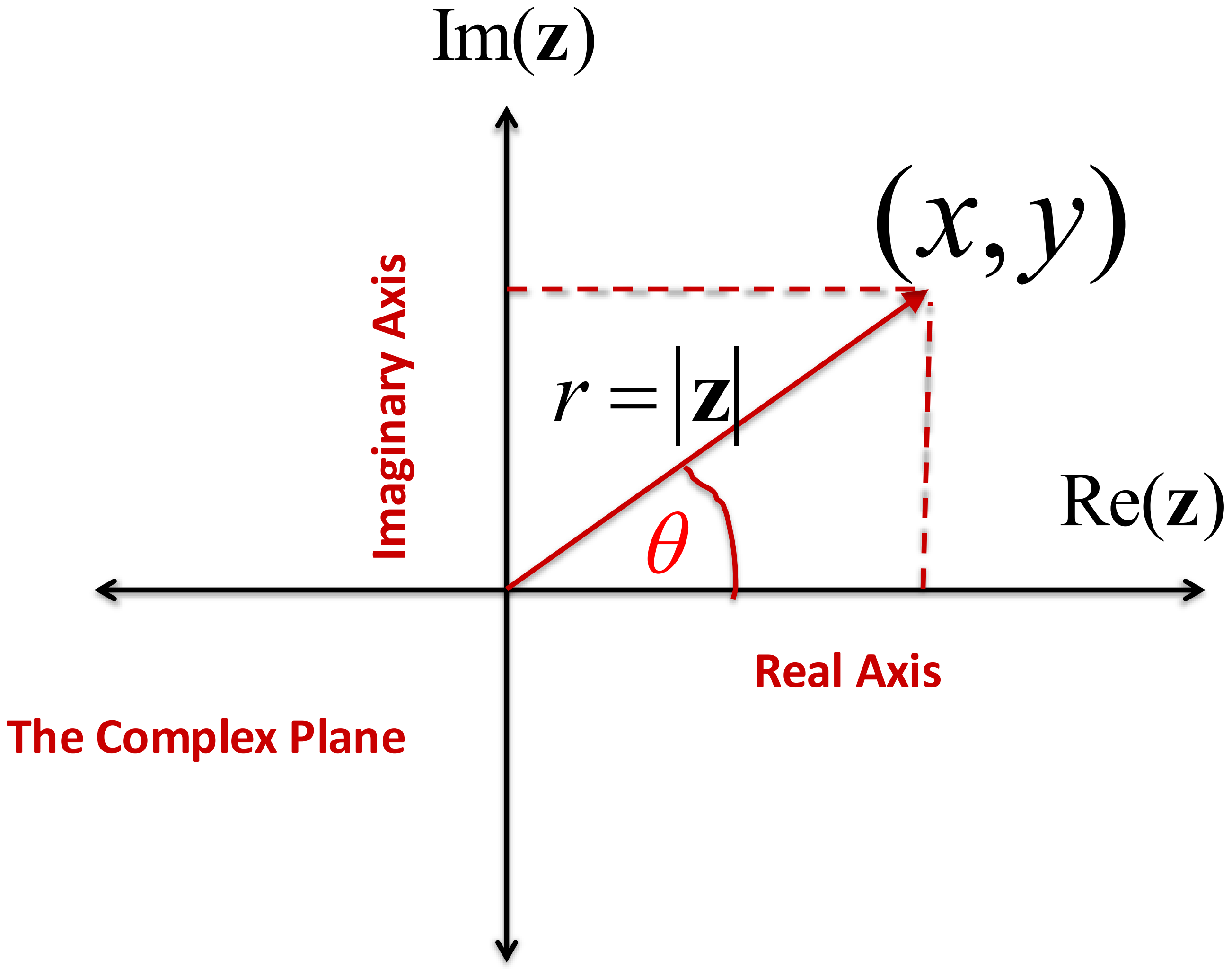
$$|\mathbf{z}| = r = \sqrt{x^2 + y^2}$$ **MAGNITUDE**

$$x = \mathrm{Re}(\mathbf{z}) = r\cos(\theta)$$

$$y = \mathrm{Im}(\mathbf{z}) = r\sin(\theta)$$

$$\mathbf{z} = r\cos(\theta) + j\,r\sin(\theta)$$

The Angle (argument)



$$\theta = \square\, \mathbf{z}$$

ANGLE IS DEFINED RELATIVE TO THE REAL AXIS

WE PREFER TO END UP WITH AN ANGLE BETWEEN -180^{O} AND + 180^{O}

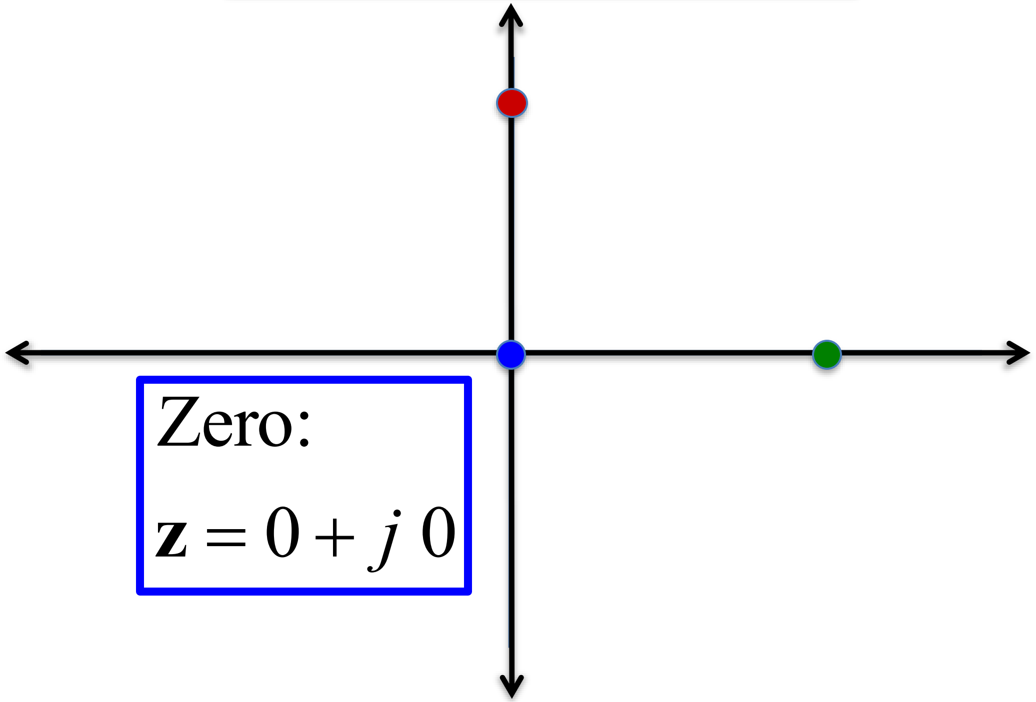
Calculating angles needs some care

MORE ON THIS SOON

Special Complex Numbers

Purely Imaginary:

$\mathbf{z} = 0 + j\,y, \quad \angle \mathbf{z} = \pm 90^{o}$



Purely Real:

$\mathbf{z} = x + j\,0, \quad \angle \mathbf{z} = 0$

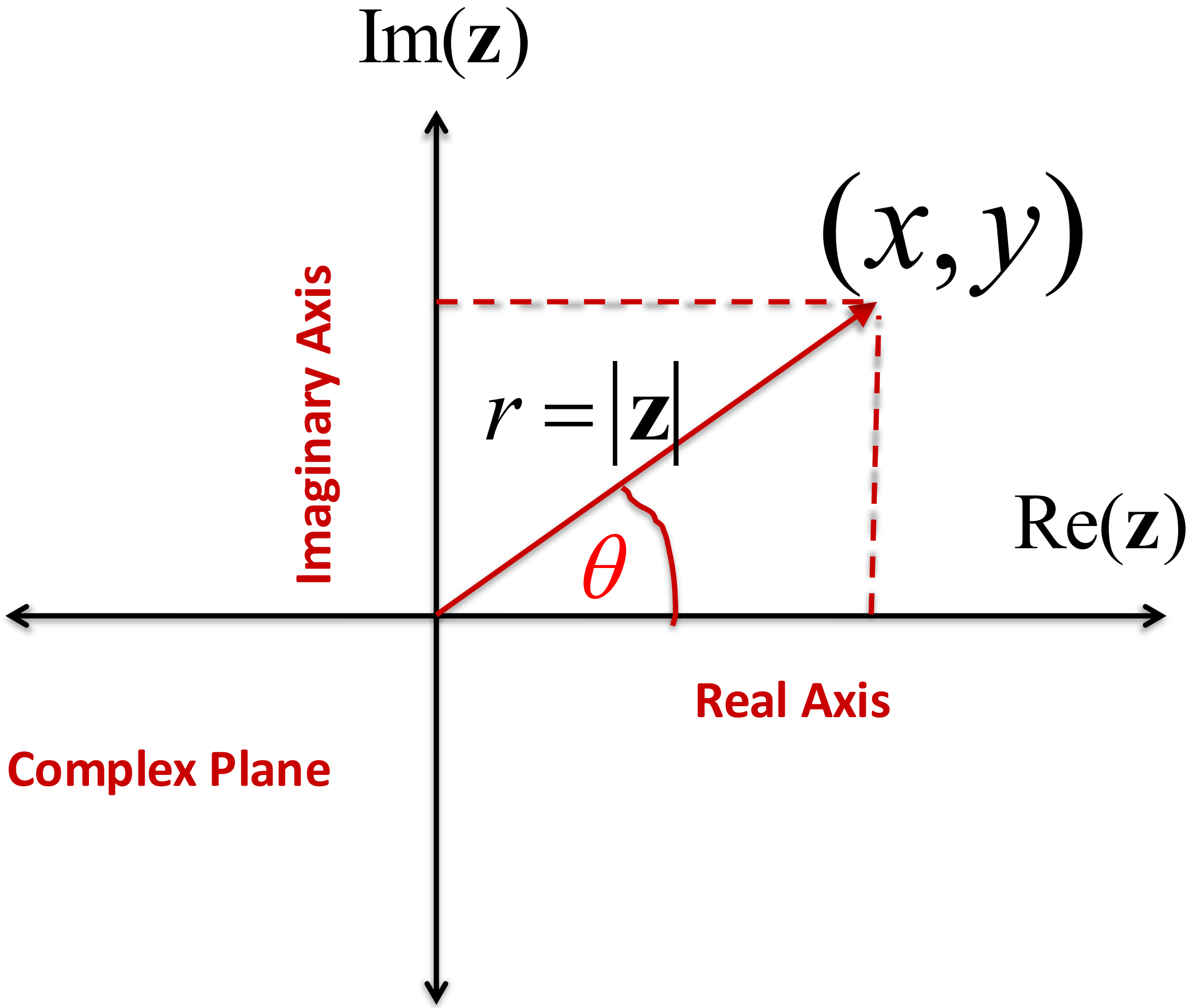
Zero:

$\mathbf{z} = 0 + j\,0$

THE ANGLE IS <u>UNDEFINED</u> FOR ZERO

COMPLEX INFINITY IS A FAR AWAY POINT, WITH AN <u>UNDEFINED</u> ANGLE

Polar Form



The Complex Plane

$$\mathbf{z} = re^{j\theta}$$

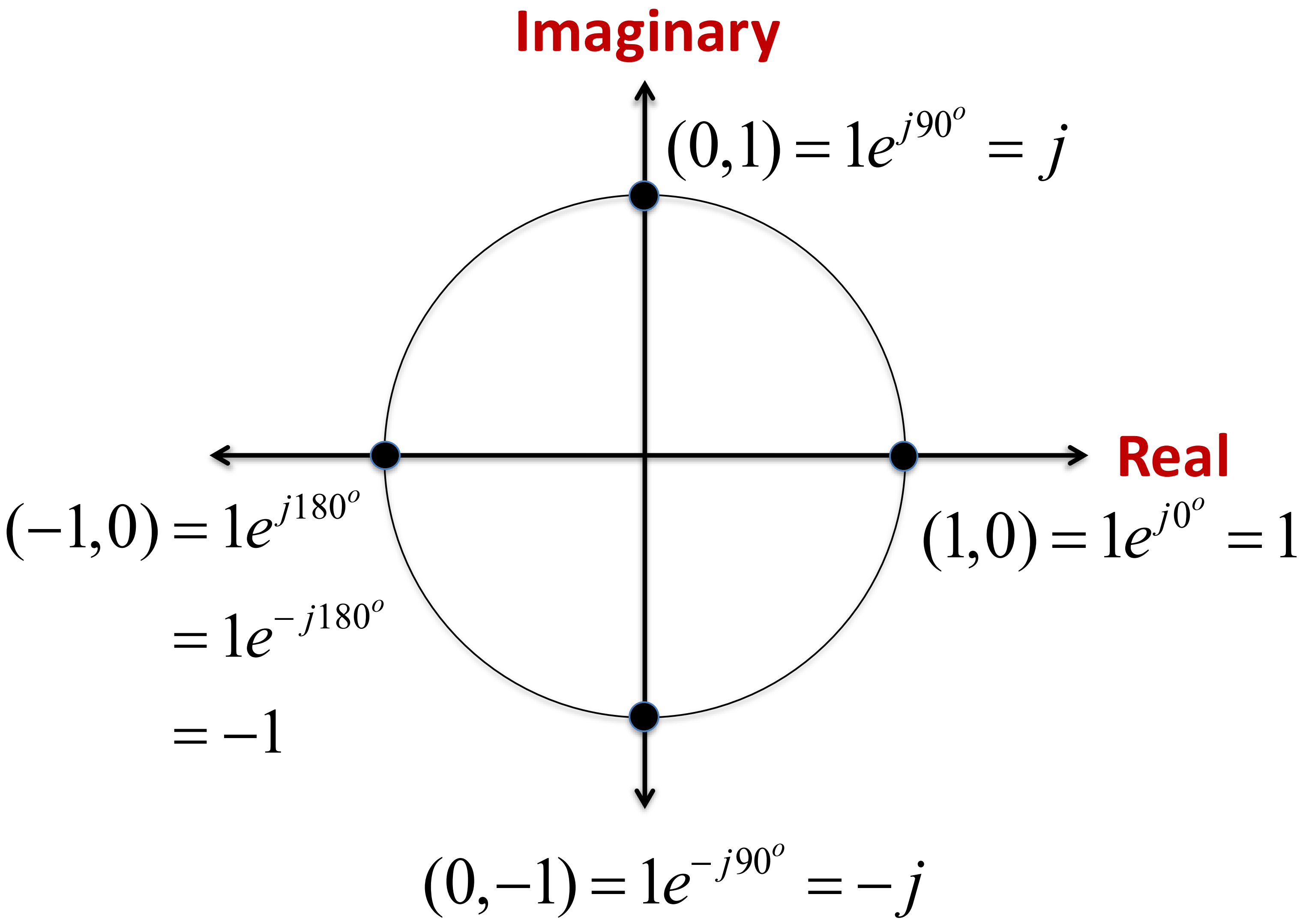
$$|\mathbf{z}| = r = \sqrt{x^2 + y^2}$$ MAGNITUDE

$$x = \mathrm{Re}(\mathbf{z}) = r\cos(\theta)$$

$$y = \mathrm{Im}(\mathbf{z}) = r\sin(\theta)$$

$$\mathbf{z} = r\cos(\theta) + j\, r\sin(\theta)$$

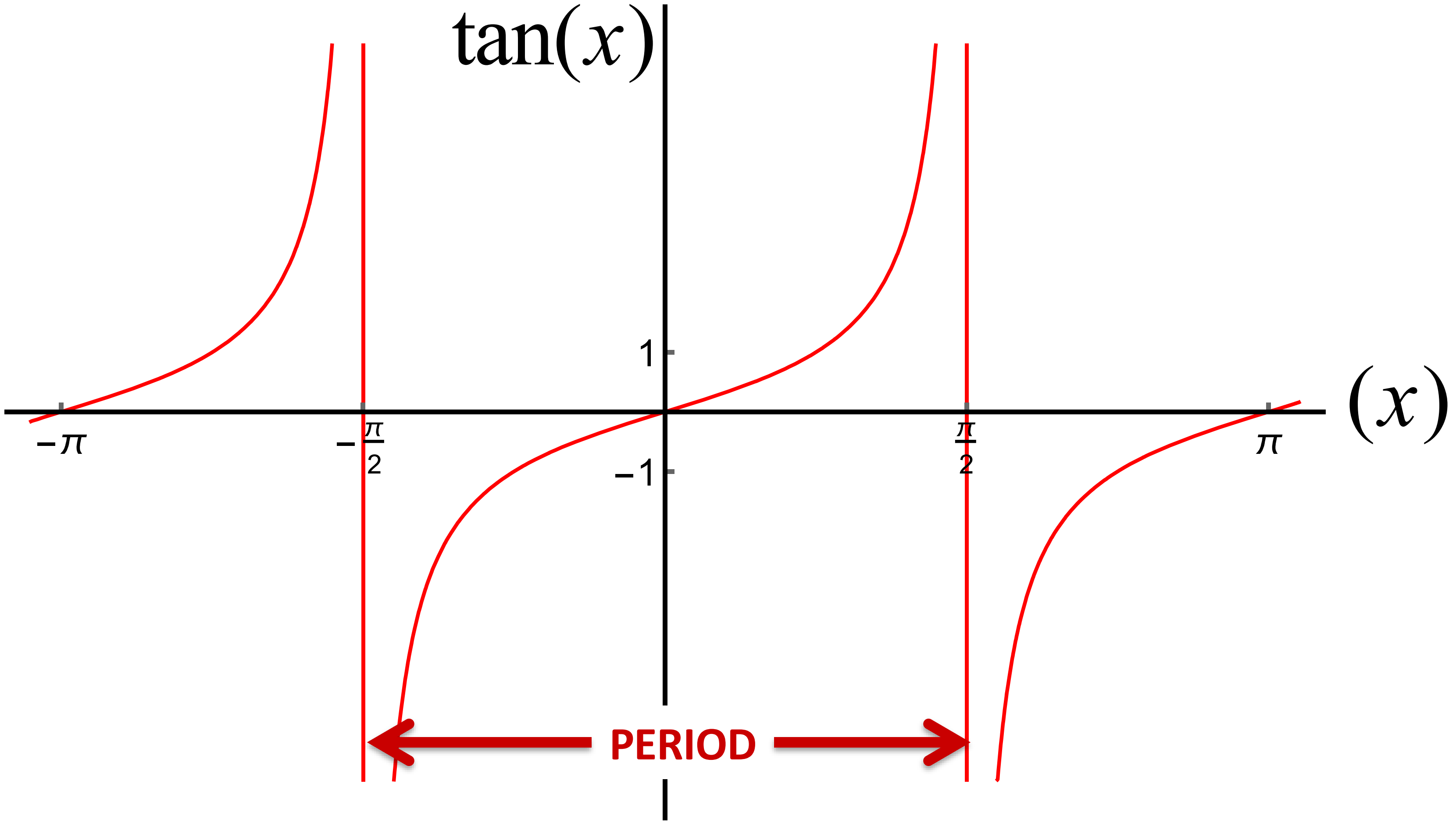
The Unit Circle



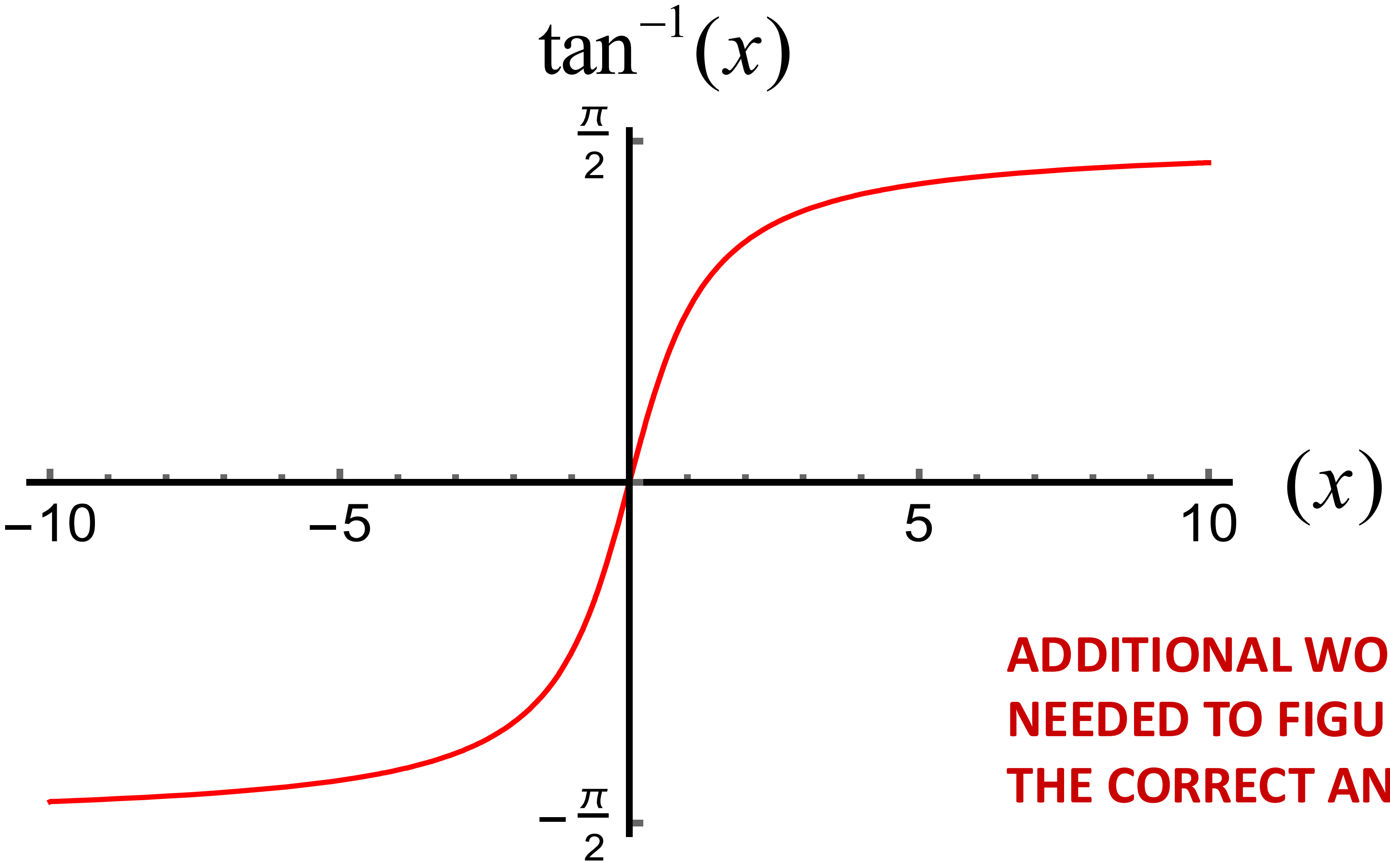
Recap: Tangent

Another periodic function, but with a period π radians

- **Cosine and sine have a period 2π radians**



Inverse Tangent



ADDITIONAL WORK IS NEEDED TO FIGURE OUT THE CORRECT ANGLE !

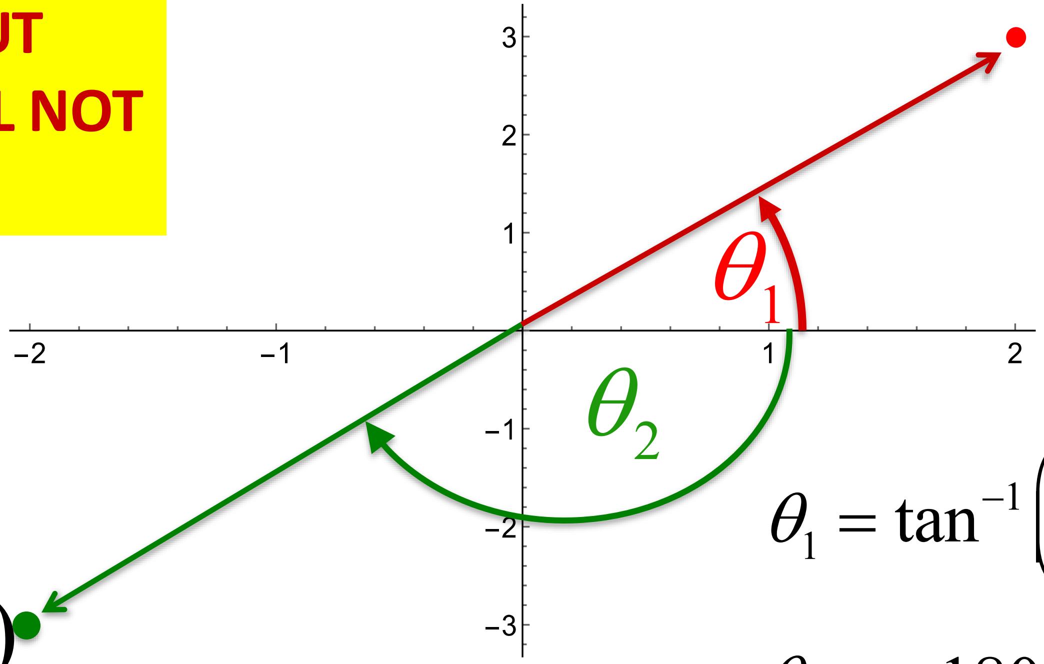
Careful with Angles!

Given a complex variable in Cartesian form (*x*, *y*), it is often necessary to convert it to the polar representation (*r*, θ)

Example:

Z_1 and Z_2 HAVE VERY DIFFERENT ANGLES, BUT THE ARCTANGENT WILL NOT TELL THEM APART

$$\mathbf{z}_1 = (2, 3)$$

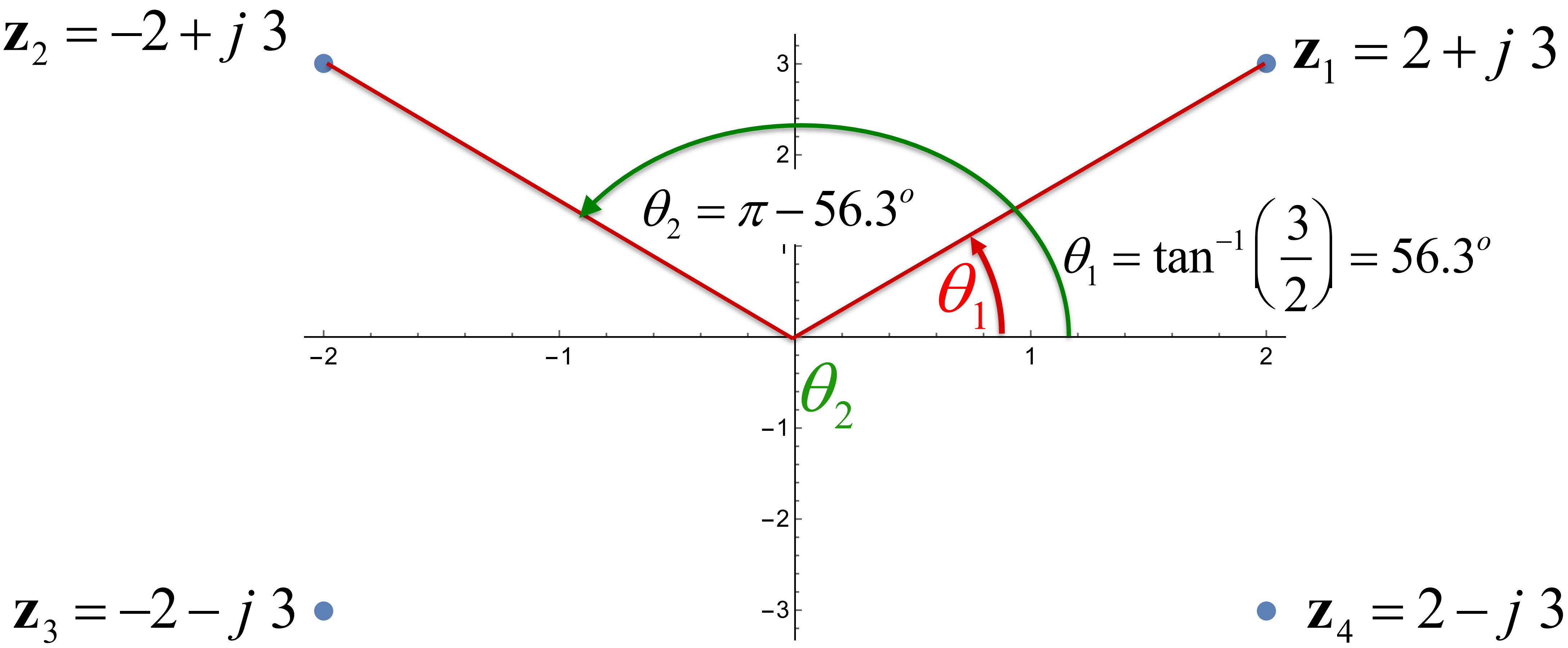


$$\mathbf{z}_2 = (-2, -3)$$

$$\theta_1 = \tan^{-1}\left(\frac{3}{2}\right) = 56.3^o$$

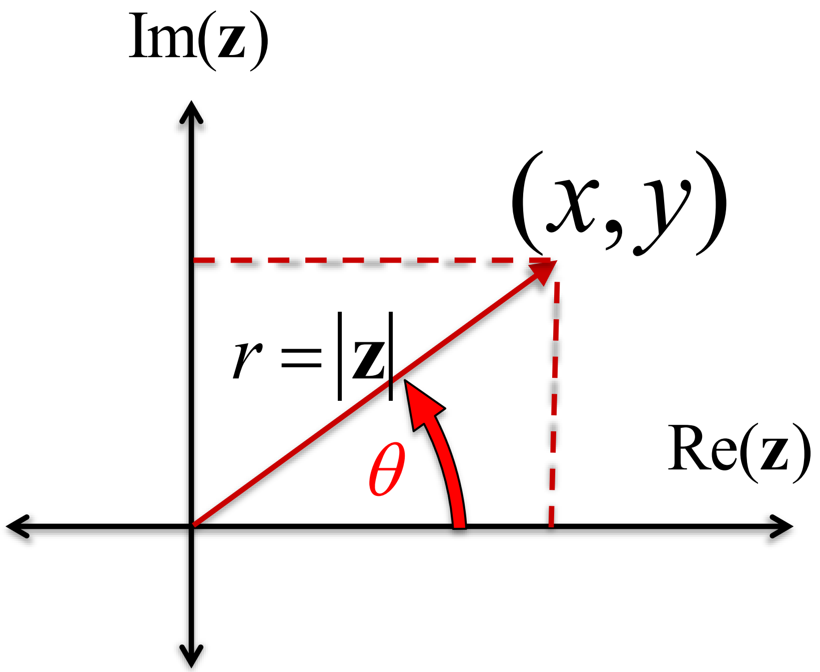
$$\theta_2 = -180^o + 56.3^o$$

Another Example



More on Angle

$$\theta = \angle \mathbf{z} = \mathrm{Arg}(x + jy) = \mathrm{atan}\,2(y, x) = \begin{cases} \arctan(\frac{y}{x}) & \text{if } x > 0 \\ \arctan(\frac{y}{x}) + \pi & \text{if } x < 0 \text{ and } y \geq 0 \\ \arctan(\frac{y}{x}) - \pi & \text{if } x < 0 \text{ and } y < 0 \\ +\frac{\pi}{2} & \text{if } x = 0 \text{ and } y > 0 \\ -\frac{\pi}{2} & \text{if } x = 0 \text{ and } y < 0 \\ \text{undefined} & \text{if } x = 0 \text{ and } y = 0 \end{cases}$$



$$\mathbf{z} = re^{j\theta}$$

Complex Algebra

$$\mathbf{z}_1 = x_1 + jy_1 \qquad \mathbf{z}_2 = x_2 + jy_2$$

Addition:

$$\mathbf{z}_1 + \mathbf{z}_2 = (x_1 + x_2) + j(y_1 + y_2)$$

Multiplication:

$$\mathbf{z}_1\mathbf{z}_2 = (x_1 + jy_1) \cdot (x_2 + jy_2)$$

$$= (x_1x_2 - y_1y_2) + j(x_1y_2 + y_1x_2)$$

JUST LIKE NORMAL ALGEBRA, WITH $j = \sqrt{-1}, \quad j^2 = -1$

The Complex Conjugate

$$\mathbf{z} = x + jy$$

Conjugate:

$$\mathbf{z}^* = x - jy$$

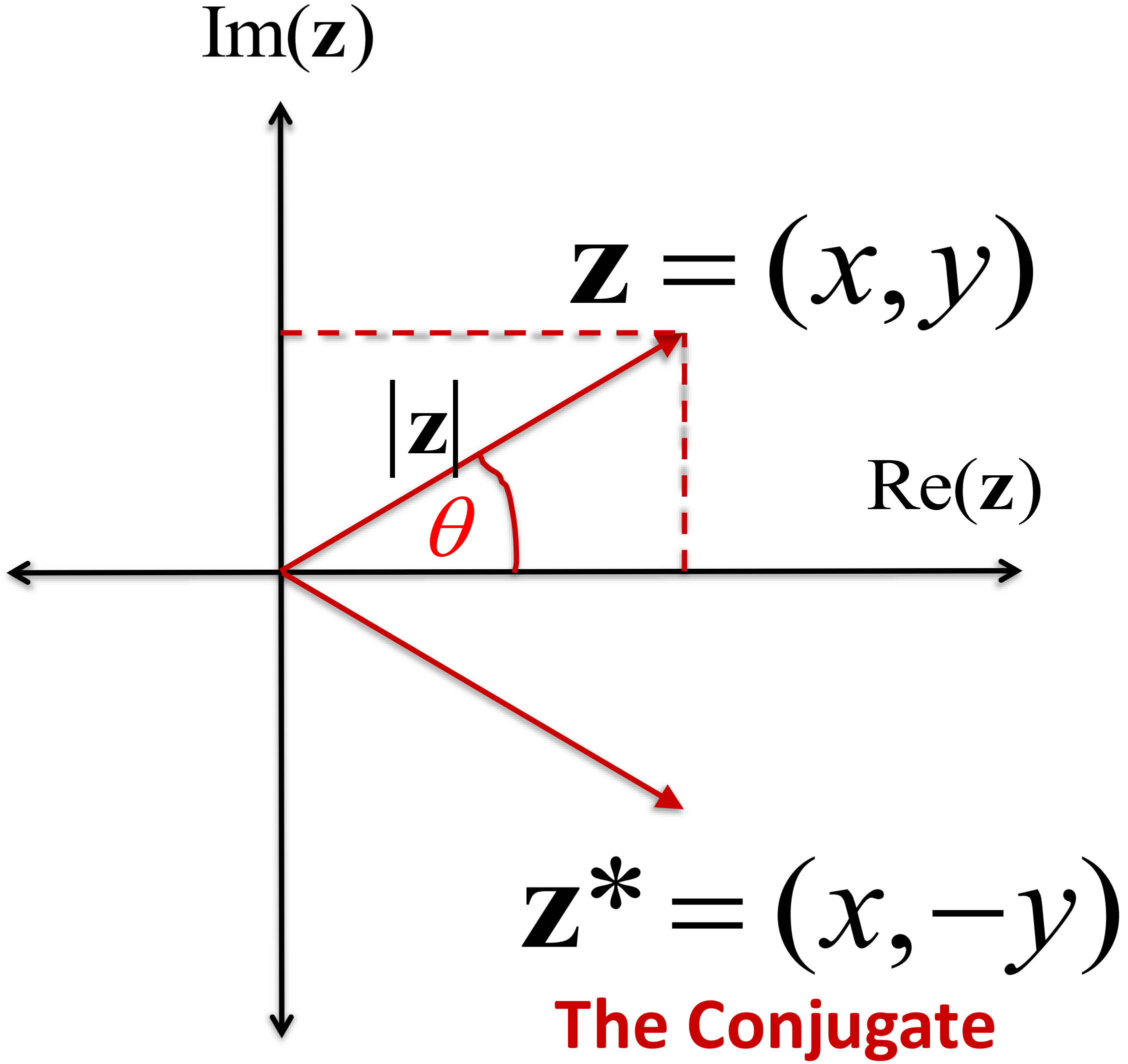
Addition of Conjugate:

$$\mathbf{z} + \mathbf{z}^* = (x + x) + j(y - y) = 2x$$

Multiplication by the Conjugate:

$$\mathbf{z}\mathbf{z}^* = (x + jy) \cdot (x - jy)$$

$$= (x^2 + y^2) + j(0) = |\mathbf{z}|^2$$



Complex Division

$$\frac{\mathbf{z}_1}{\mathbf{z}_2} = \frac{x_1 + jy_1}{x_2 + jy_2}$$

$$= \frac{x_1 + jy_1}{x_2 + jy_2} \cdot \frac{x_2 - jy_2}{x_2 - jy_2}$$

$$= \frac{(x_1 x_2 + y_1 y_2)}{(x_2^2 + y_2^2)} + j\frac{(x_2 y_1 - x_1 y_2)}{(x_2^2 + y_2^2)}$$

WITH DIVISION, WE NEED TO ALWAYS EXPLICITLY MAKE SURE THAT THE DENOMINATOR IS NOT ZERO

WHEN THE DENOMINATOR IS ZERO, THE RESULTS ARE UNDEFINED

The Reciprocal

Reciprocal:

$$\frac{1}{\mathbf{z}} = \frac{1}{x + jy}$$

$$= \frac{1}{x + jy} \cdot \frac{x - jy}{x - jy} = \frac{x}{(x^2 + y^2)} - j\frac{y}{(x^2 + y^2)}$$

SAME STANDARD TRICK

AS WITH DIVISION, <u>EXPLICITLY</u> MAKE SURE THAT THE DENOMINATOR IS NOT ZERO

WHEN THE DENOMINATOR IS ZERO, THE RESULTS ARE UNDEFINED (i.e., MEANINGLESS), AND WE NEED DO SOMETHING DIFFERENT

Complex Algebra in Polar Coordinates

$$\mathbf{z}_1 = r_1 e^{j\theta_1} \qquad \mathbf{z}_2 = r_2 e^{j\theta_2}$$

Conjugates:

$$\mathbf{z}_1{}^* = r_1 e^{-j\theta_1} \qquad \mathbf{z}_2{}^* = r_2 e^{-j\theta_2}$$

PROVIDED *r* IS NON-ZERO

Addition:

$$\mathbf{z}_1 + \mathbf{z}_2 = (r_1 \cos\theta_1 + r_2 \cos\theta_2) + j(r_1 \sin\theta_1 + r_2 \sin\theta_2)$$

Multiplication:

$$\mathbf{z}_1 \mathbf{z}_2 = r_1 r_2 e^{j(\theta_1 + \theta_2)}$$

Division:

$$\frac{\mathbf{z}_1}{\mathbf{z}_2} = \frac{r_1 e^{j\theta_1}}{r_2 e^{j\theta_2}} = \frac{r_1}{r_2} e^{j(\theta_1 - \theta_2)}$$

PROVIDED THE DENOMINATOR IS NON-ZERO

Magnitude & Angle of a Ratio

$$\left|\frac{x_1 + jy_1}{x_2 + jy_2}\right| = \frac{|x_1 + jy_1|}{|x_2 + jy_2|} = \frac{r_1}{r_2} = \frac{\sqrt{x_1^2 + y_1^2}}{\sqrt{x_2^2 + y_2^2}}$$

WHEN THE DENOMINATOR IS ZERO, THE RESULTS ARE UNDEFINED

$$\angle\left(\frac{x_1 + jy_1}{x_2 + jy_2}\right) = \angle\,(x_1 + jy_1) - \angle(x_1 + jy_1)$$

Magnitude & Angle of a Product

$$|(x_1+jy_1)(x_2+jy_2)| = |x_1+jy_1||x_2+jy_2|$$

$$r_1 r_2 = \sqrt{x_1^2+y_1^2}\sqrt{x_2^2+y_2^2}$$

$$\angle(x_1+jy_1)(x_2+jy_2) = \angle(x_1+jy_1) + \angle(x_1+jy_1)$$

Complex Exponential Signal

Sometimes, the amplitude is a complex quantity. This is not a showstopper: just complete the complex multiplication and simplify

$$x(t) = \underline{\mathbf{A}} e^{j(\omega_0 t+\varphi)} \qquad \mathbf{A} = |A| e^{j\theta}$$

Complex amplitude

$$\square \quad x(t) = |A| e^{j\theta} \,\square\, e^{j(\omega_0 t+\varphi)} = |A| e^{j(\omega_0 t+\theta+\varphi)}$$

This is a periodic signal with complex frequency ω_0

The frequency is the number that multiplies jt

The phase is the number that multiplies j **only**

Terms without j **(real terms) fall into the magnitude** $|A|$

Complex Exponential Signals

$$\boldsymbol{x_1}(t) = A_1 e^{j(\omega_1 t + \varphi_1)}$$

$$\boldsymbol{x_2}(t) = A_2 e^{j(\omega_2 t + \varphi_2)}$$

$$\boldsymbol{x_1}(t)\boldsymbol{x_2}(t) = A_1 e^{j(\omega_1 t + \varphi_1)} A_2 e^{j(\omega_2 t + \varphi_2)}$$

$$= A_1\ A_2\ e^{j(\omega_1 + \omega_2)t + (\varphi_1 + \varphi_2)}$$

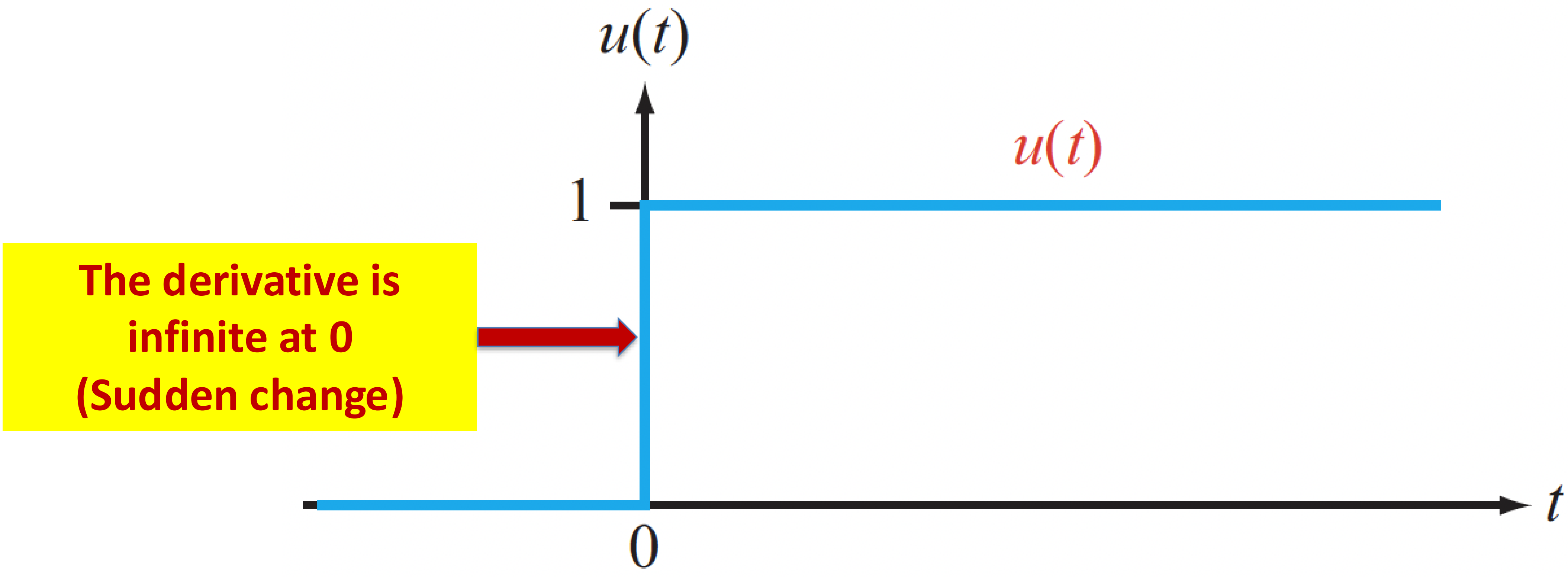
This is a periodic signal with frequency $(\omega_1 + \omega_2)$

The phase is $(\varphi_1 + \varphi_2)$

The amplitude is $A_1\ A_2$

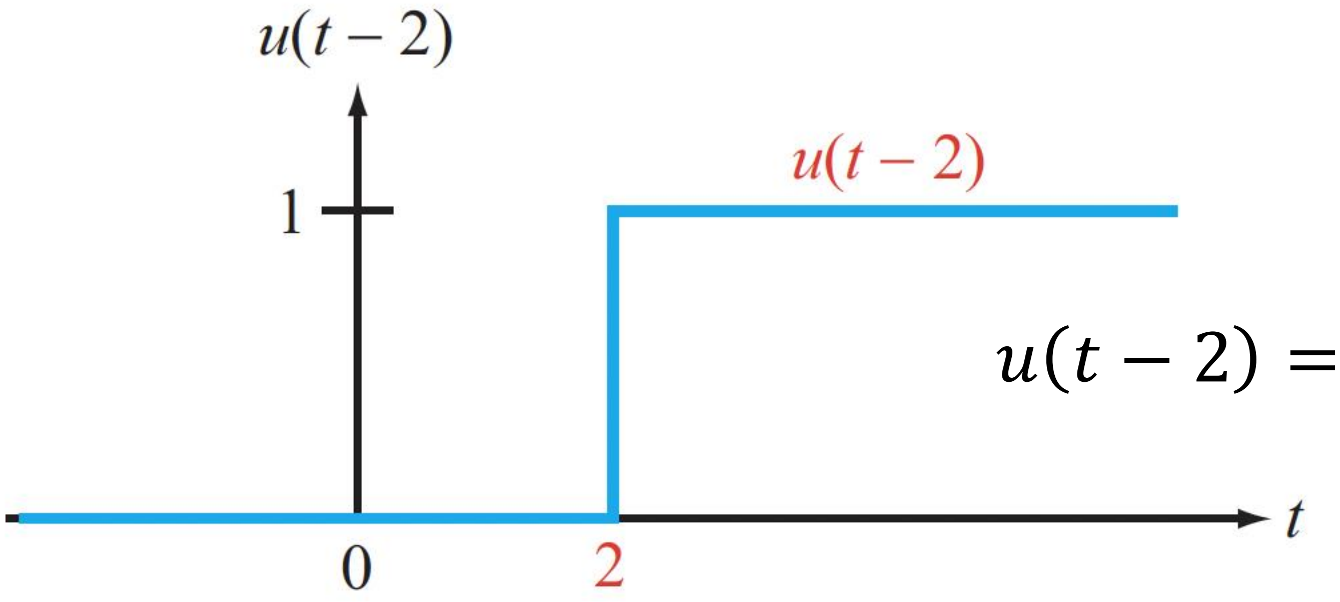
Non-periodic Waveforms

The Unit Step: Like turning ON a switch

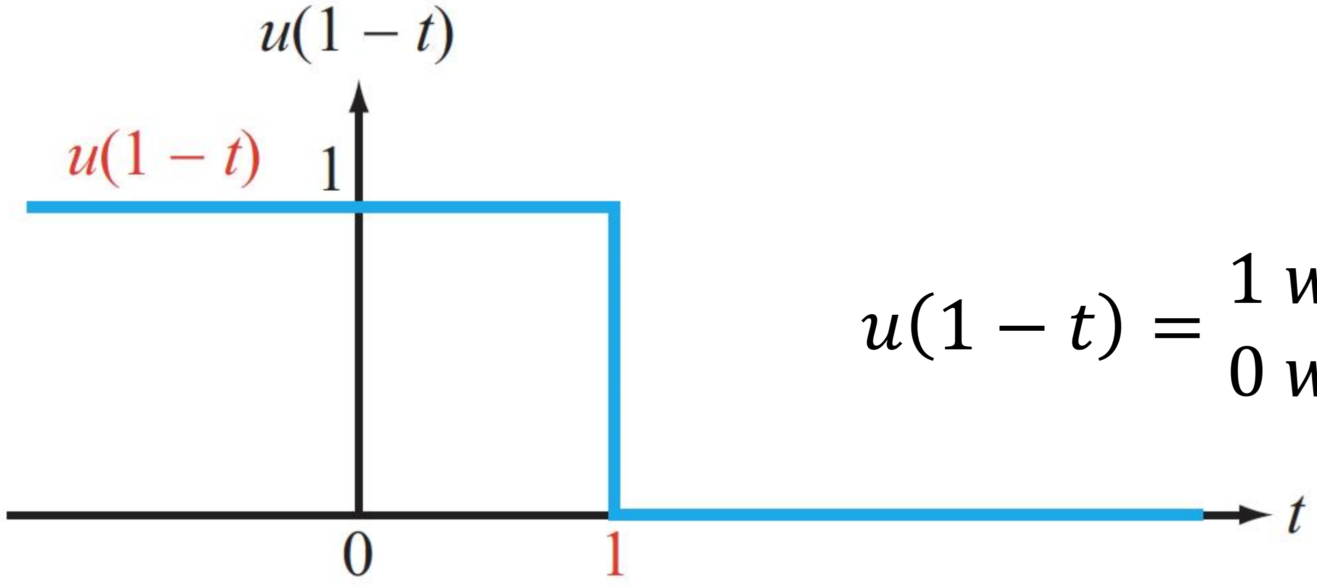


Can you sketch $u(-t)$?

Shifted Steps



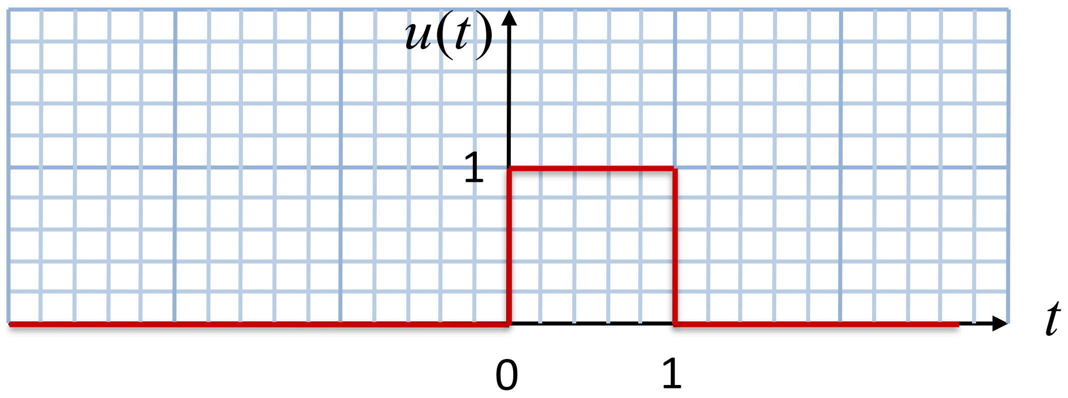
$$u(t-2) = \begin{cases} 1\ when\ (t-2) > 0, or\ t > 2 \\ 0\ when\ (t-2) < 0\ or\ t < 2 \end{cases}$$



$$u(1-t) = \begin{cases} 1\ when\ (1-t) > 0\ or\ t < 1 \\ 0\ when\ (1-t) < 0\ or\ t > 1 \end{cases}$$

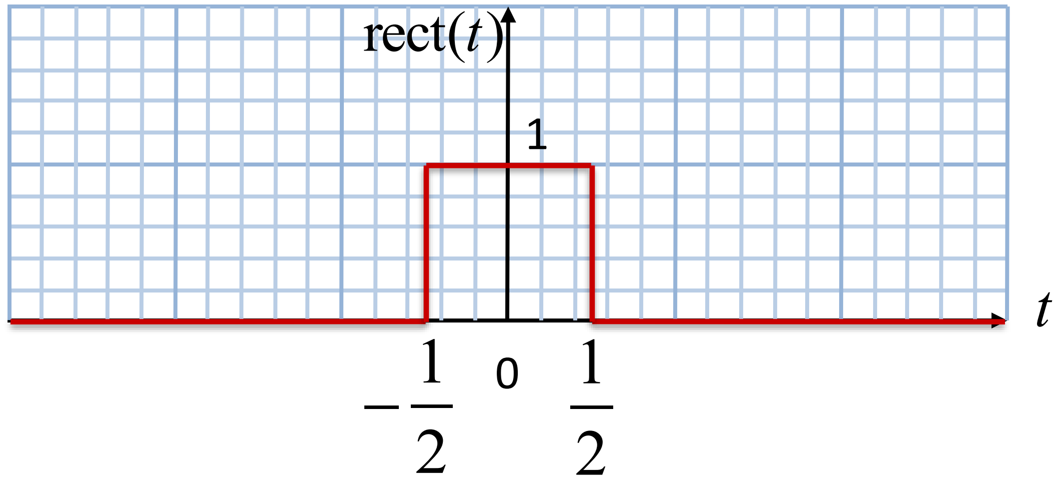
The Pulse & Unit Rectangle

$$p(t) = u(t) - u(t-1)$$



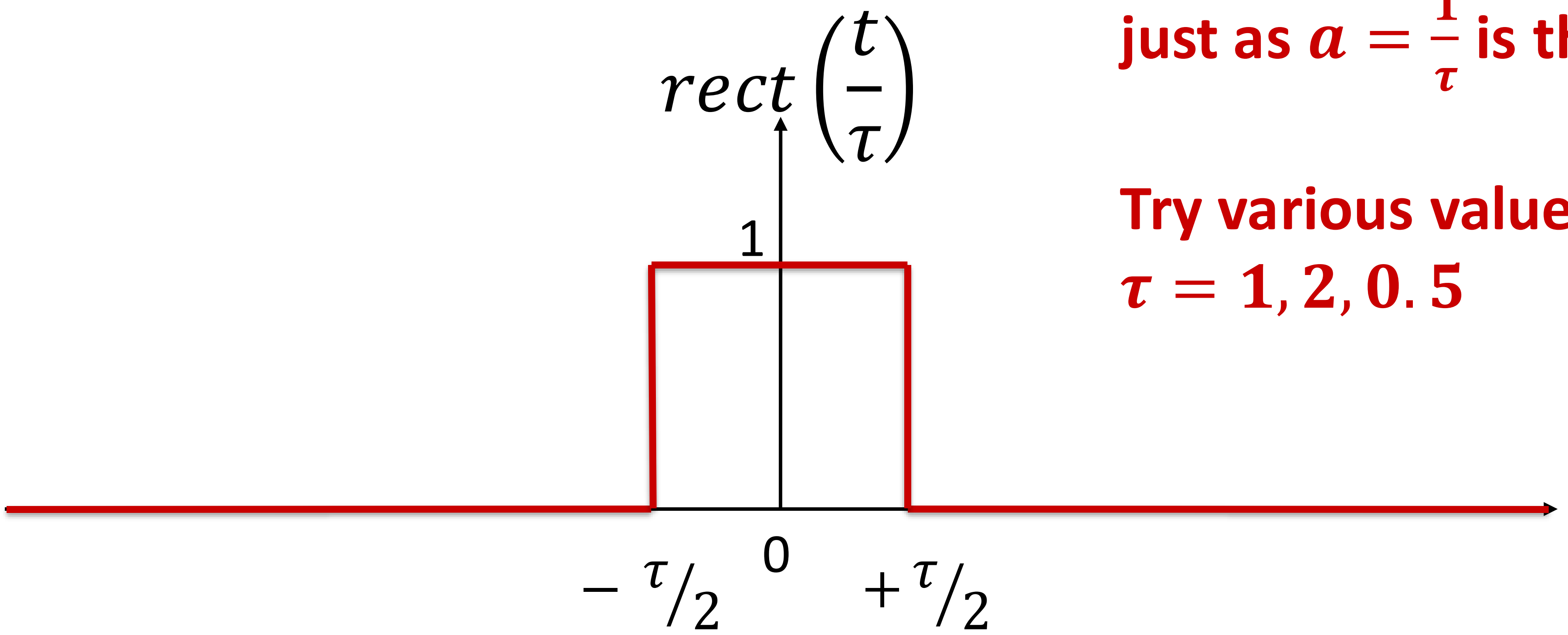
The Unit Pulse

$$\mathrm{rect}(t) = u(t+\frac{1}{2}) - u(t-\frac{1}{2})$$



The Unit Rectangle

Setting the width of a Pulse



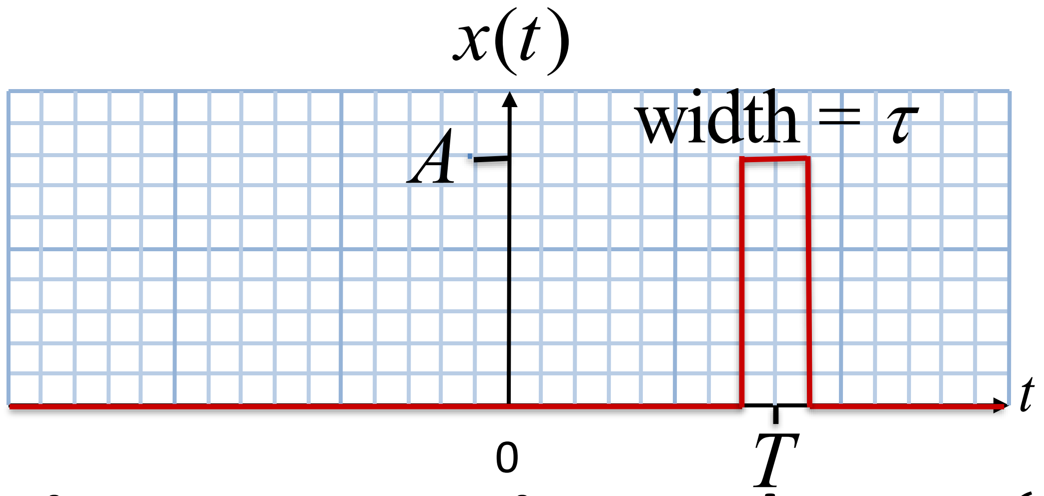
$width = \tau$

Think of τ as the "playback time" just as $a = \frac{1}{\tau}$ is the "playback speed".

Try various values of τ , for example, $\tau = 1, 2, 0.5$

Exercise

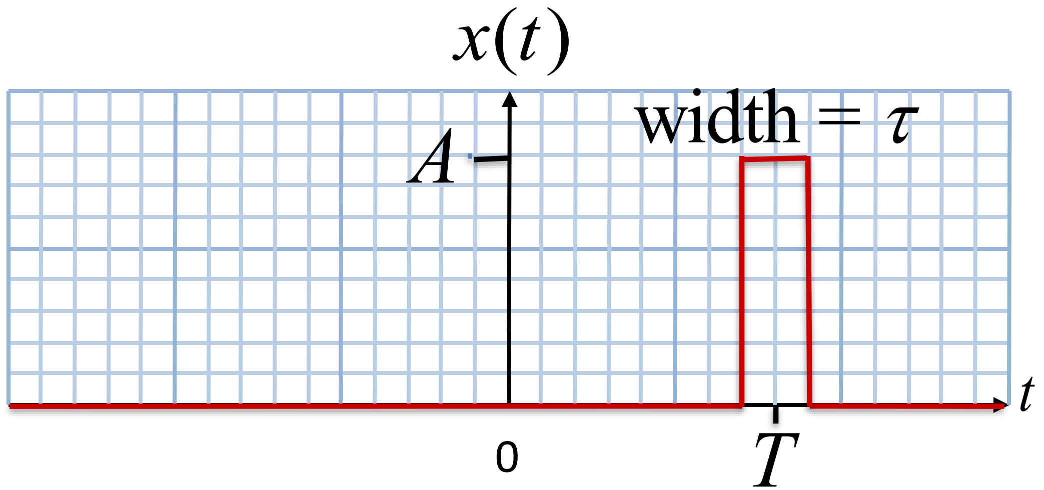
Using signal transformations, write a formula for the following signal in terms of the rect() function:



<u>Suggestion</u>: Do this in two steps: time scale $rect(t)$, and then shift right

Exercise

Using signal transformations, write a formula for the following signal in terms of the rect() function:



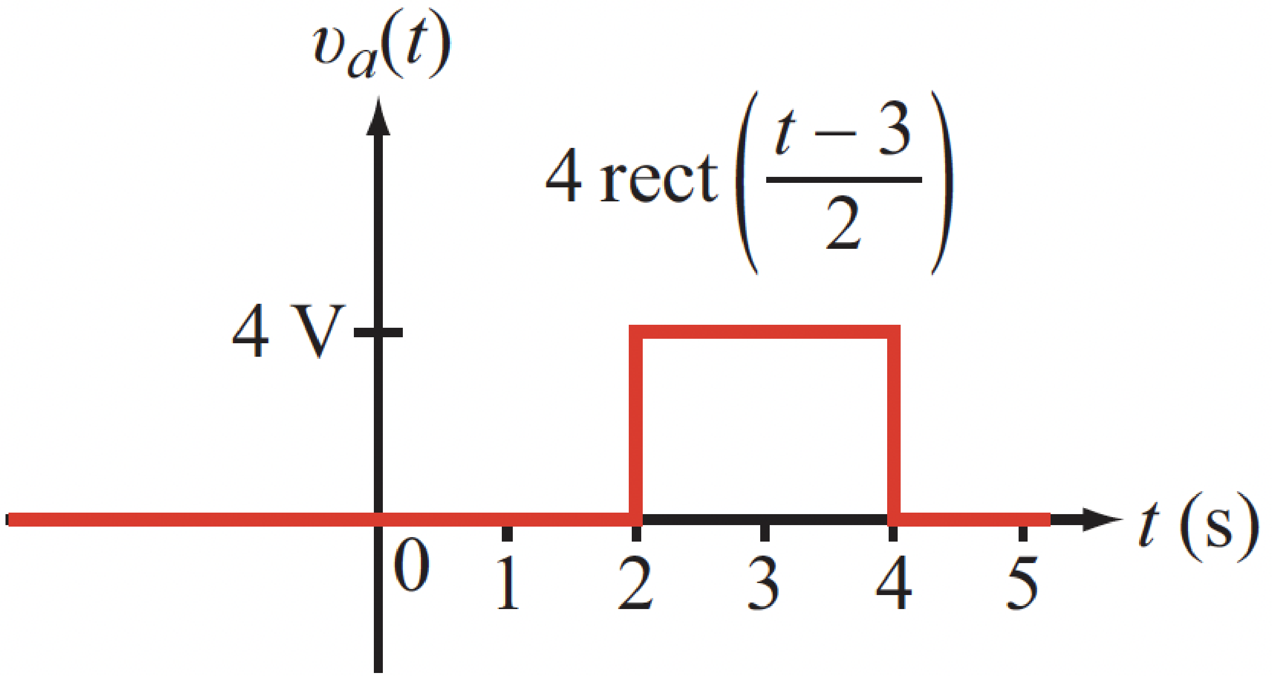
<u>Suggestion</u>: Do this in two steps: time scale $rect(t)$, and then shift right

<u>Correct Answer</u>:

$$x(t) = A\,\mathrm{rect}\left(\frac{t-T}{\tau}\right)$$

Example

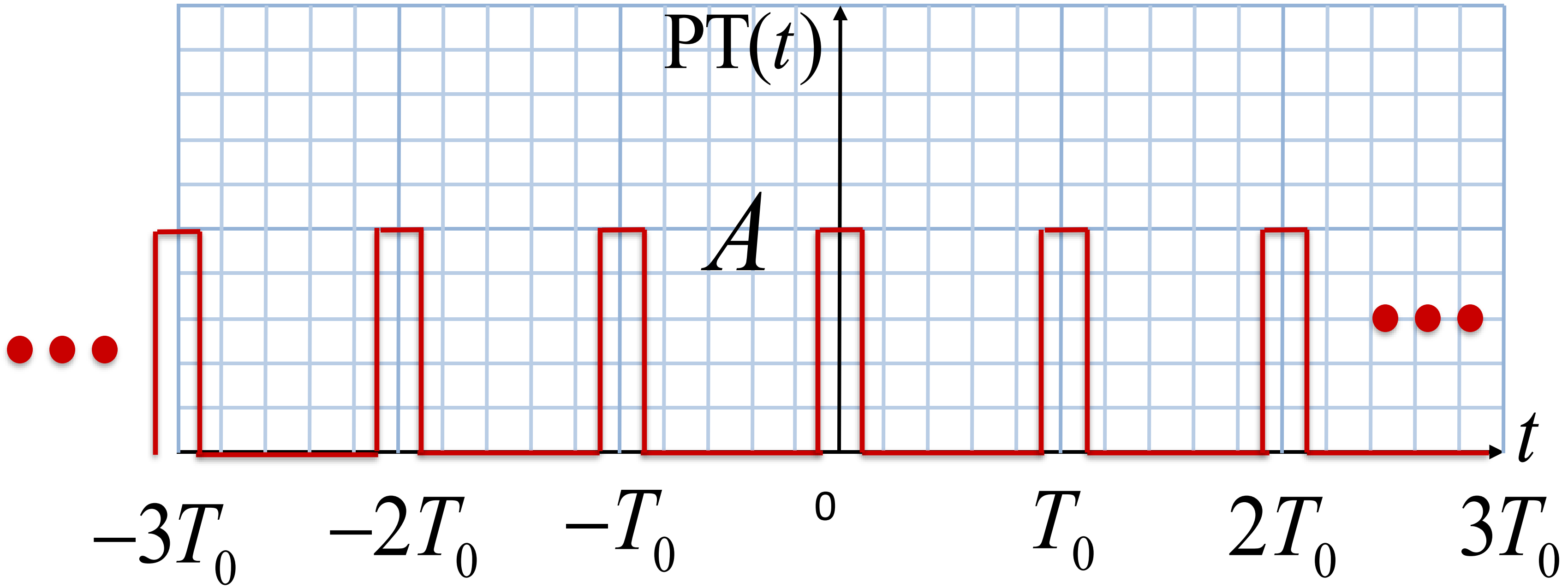
Write a formula for the following waveform



Correct Answer:

$$v_a(t) = 4u(t-2) - 4u(t-4)$$

A Common Periodic Signal: The Pulse Train



Related Terms:

- Amplitude A
- Pulse Width τ
- Time Period T_0
- Duty Cycle $\frac{\tau}{T_0}$
- Frequency $\frac{1}{T_0}$

$$PT(t) = A \sum_{n=-\infty}^{n=\infty} \text{rect}\left(\frac{t - nT_0}{\tau}\right)$$

CAN YOU THINK OF PRACTICAL USES FOR THIS SIGNAL?

Plot These Waveforms Now

$$y_1(t) = 2u(t+1)$$

$$y_2(t) = 2u(t-1)$$

$$y_3(t) = 2u(t) - u(t-1)$$

$$y_4(t) = u(t) + 2u(t-1)$$

Summary

We looked at periodic signals

- This topic is fundamental
- It requires trigonometry, complex algebra, and Euler's formula

We reviewed some essential math today

- You need to do a lot more on your own
  - Many free PDF textbooks online, and on the Library's Reference Collection
  - Worked examples posted on course Blackboard site
- Watch the video lectures that we prepared for you
- Make it a habit to work problems from the textbook every single day to keep your math skills in good condition

We looked at the simplest non-periodic signals

- The unit step, the pulse, and the rectangle
- These basic signals are building blocks for complicated signals

For the Next Class

Do the Zybooks exercises in Sections 1-3 – 1-4, and make sure you can do them unaided

Study Sections 1.4 – 1.5 on the Zybook, and do the Activities

How to Study:

1. **Read the assigned sections to understand the basic concepts, and try to memorize the concepts**
2. **Attempt the in-chapter exercises and practice until you can do them fully on your own without notes, like in a test**
3. **Write down your questions and bring them to class**